Drone Frame



Category: **Drones & Aviation** Type: **Short SOP** Priority: **Medium** 3 labs

Covers these lab items: Drone base; Foldable Air Frame – Aluminium & Glass Fibre

What it is used for

To assemble and inspect a drone frame before installing electronics and propulsion components. To ensure the frame is correctly aligned, securely fastened, and suitable for the intended drone configuration.

Main parts

- Central frame or body plate
- Arms or booms
- Motor mounting points
- Landing gear, if fitted
- Fasteners and spacers
- Foldable hinges or locking joints, if fitted

Before you start

1. Confirm the frame size and material are compatible with the planned drone components.
2. Inspect the frame, arms, hinges, and mounting points for cracks or deformation.
3. Prepare the correct fasteners and tools.
4. Work on a clean, stable surface with power disconnected from all electronics.
5. Review the frame assembly instructions.

How to use

1. Lay out all frame parts and verify them against the assembly instructions.
2. Attach the arms to the central frame in the specified orientation.
3. Fit the specified fasteners and tighten them evenly to the recommended requirement.
4. Check that motor mounts and electronic mounting points are aligned.
5. For foldable frames, test each hinge and locking mechanism.
6. Inspect the frame for twisting, loose parts, and sharp edges.
7. Install electronics only after the frame passes inspection.

⚠ Safety

- Never power motors while assembling or adjusting the frame.
- Use the specified fasteners and avoid overtightening composite parts.
- Wear eye protection when drilling or cutting is explicitly required.
- Do not fly with cracked arms, loose joints, or damaged locking mechanisms.

Care & storage

- Remove dust and debris after assembly.
- Inspect joints and fasteners before each flight.
- Keep the frame dry and protected from impacts.
- Replace cracked or deformed parts before use.

Common mistakes to avoid

- Installing arms in the wrong orientation.
- Using incorrect fasteners or excessive tightening force.
- Ignoring a loose foldable hinge.
- Installing electronics on an unchecked or damaged frame.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

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